

Where Does Certainty Live in a Transformer?

Or: Why Verbal Confidence is a Trap, Sampling Tells You Everything, and Factual Memory is Instant While Reasoning Takes Depth

Alignment & MechInterp

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EPISTEMIC STATUS:

High confidence in empirical results across Qwen2.5 (0.5B to 7B, 30 cells, 588 probe fits, 4.778 GPU-hours on Molab's RTX PRO 6000 Blackwell). All gates and hypotheses were pre-registered in `PLAN.md` before execution, and final artifacts are bound to commit SHAs (coarse: `7570841`, fine: `a4608a8`). Negative control (`$p_0$`) properly fired. Moderate confidence that the ~25% depth gap between associative memory lookup and multi-step reasoning is an architectural invariant of decoder-only transformers. High confidence that single-prompt verbal confidence is fundamentally broken as a measure of model belief.

1. The Illusion of the Unified Dial

When you prompt a modern language model with a challenging question and ask, *"How confident are you in this answer from 0% to 100%?"*, it might write back:

"The answer is 42. I am 95% confident in this result."

It is tempting—almost irresistible—to imagine that inside the transformer weights, there is a single internal scalar dial labeled **"Epistemic Confidence"**. Under this mental model, the model evaluates its uncertainty, reads off the dial, and prints "95%". If this were true, calibration would be simple: calibrate the dial once, and you can reliably route queries to humans when the dial reads low.

Over the past two months, our team conducted a massive, pre-registered audit across 30 experimental cells (5 model sizes from Qwen2.5 0.5B to 7B across 6 factual retrieval and multi-step reasoning tasks). We measured three competing channels simultaneously on identical items:

1. **The Verbalizer (Words):** What the model states in tokens (0–100%, qualitative verbal bins, or betting stakes).
2. **The Residual Stream (Weights / Activations):** Linear logistic probes trained on hidden states during an unperturbed prefill forward pass, tapped at the *last non-padded prompt token* across 21 layer depth percentiles.

3. **The Rollouts (Behavioral Consensus):** Generating 10 stochastic samples at temperature $T = 0.7$ and measuring semantic entropy across answer clusters.

The headline finding can be stated bluntly: **The unified dial does not exist.**

The three signals routinely disagree, decouple, and deceive. A model stating "95% confidence" is often hallucinating with swagger; a model with pristine calibration curves is often pulling a statistical sleight-of-hand; and the layer where a model "knows" whether it will get an answer right depends radically on whether it is recalling a fact or computing a logical deduction.

2. The Climatology Trick: How Verbal Confidence Cheats on Calibration

In the machine learning literature, papers frequently celebrate verbal calibration. They plot Expected Calibration Error (ECE) and show curves that hug the diagonal: *"Look! Our model has an ECE of only 0.035! It is well-calibrated!"*

When we ran our pooled evaluation, we saw the exact same thing:

- Verbal ECE = **0.0356**
- Internal Probe ECE = **0.0368**
- Behavioral Consensus ECE = **0.0427**

A naive reading says: *all three signals are equally calibrated; verbal statements are just as good as expensive multi-sample rollouts.*

This is a complete illusion.

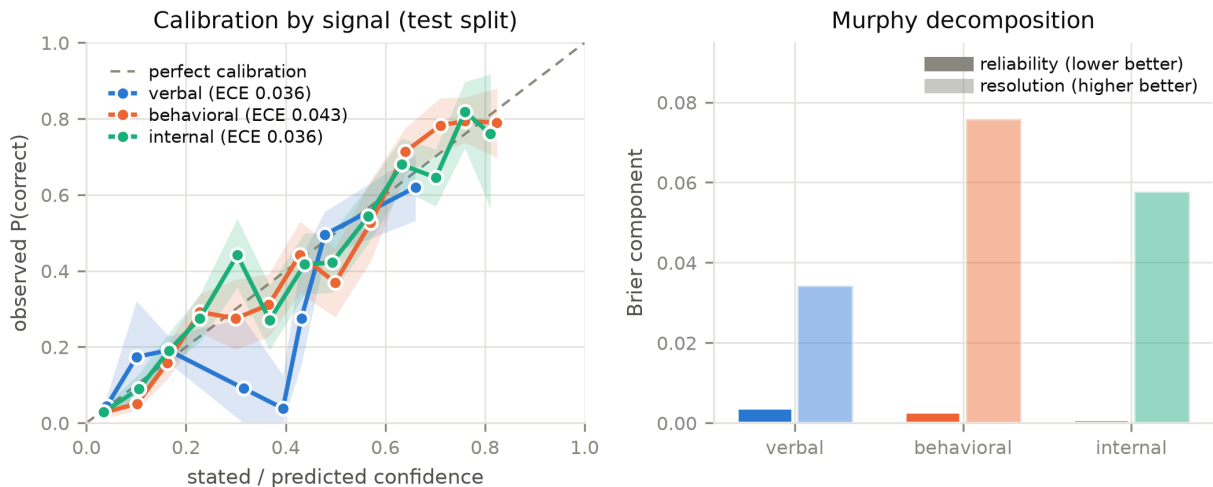


Figure 1: The ECE Illusion Unveiled. *Left:* Calibration curves for verbal, behavioral, and internal signals on the test split (points with $N < 10$ pruned). All three hug the diagonal. *Right:* Murphy's Brier score decomposition. While all three signals have negligible calibration error (dark solid bars), behavioral consensus achieves more than double the discriminative resolution (light tall bars) of verbal confidence.

Enter Allan Murphy (1973)

In meteorology, there is a famous trick known as the **Climatology Forecast**. If you are forecasting rain in London, and you know the historical base rate of rain is 28%, you can simply predict a flat 28% *probability of rain every single day of the year*.

If you evaluate your forecast using ECE, your score is virtually zero! You are "perfectly calibrated"! But your forecast is completely useless: you cannot tell someone whether they should bring an umbrella today.

Fifty years ago, statistician Allan Murphy proved that the Brier score (mean squared error of probabilities) decomposes into three orthogonal terms:

$$\text{Brier Score} = \underbrace{\text{Reliability}}_{\text{Miscalibration Error (lower is better)}} - \underbrace{\text{Resolution}}_{\text{Sorting Power (higher is better)}} + \underbrace{\text{Uncertainty}}_{\text{Task Base Rate}}$$

- **Reliability** penalizes you when your stated 80% bin has an empirical accuracy of 60%.
- **Resolution** rewards you for sorting questions into distinct buckets that deviate from the background base rate. *Resolution is what tells you whether the model actually knows which answers are right vs. wrong.*

The Strict Complete-Case Test (\$N = 842\$) & Binning Sensitivity

When we restricted our analysis to the strict complete-case subset (\$N = 842\$ questions where all three signals were evaluated on the exact same models and items, with base rate $\bar{y} = 0.2815$), the truth became unmistakable:

Confidence Signal	ECE (15 Bins) (↓)	ECE (10 Bins) (↓)	Reliability Error (↓)	Murphy Resolution (↑)	Brier Score (↓)
Verbalized (Words)	0.0366	0.0326	0.00171	0.03160	0.1574
Internal Probe (Weights)	0.0499	0.0338	0.00219	0.05154	0.1557
Behavioral Consensus (Rollouts)	0.0704	0.0680	0.00808	0.06371	0.1484

Notice what happened across both binning specifications:

1. **Behavioral consensus actually loses on ECE:** Sampling 10 rollouts at $T = 0.7$ polarizes the consensus towards 0 or 1, incurring a small calibration penalty (ECE = 0.070 vs 0.037 under 15 bins; 0.068 vs 0.033 under 10 bins).
2. **HOWEVER, Behavioral consensus achieves more than DOUBLE the resolution of verbal statements (0.06371 vs. 0.03160, a 2.02× gain).** It sorts right answers from wrong answers with twice the discriminative power.
3. **Verbal confidence achieves low ECE by playing the Climatology Trick:** On difficult reasoning benchmarks, calibrated verbal confidence never exceeds 0.50 in 8 of the 9 evaluated cells. It hugs the base rate, takes no epistemic risks, and provides essentially zero sorting ability.

3. The Speed of Knowing: Where Does Certainty Form?

Now let's turn to mechanistic interpretability. If we train linear probes on the hidden representations of the model's residual stream during prompt prefill, at what layer does the model "know" whether it will get the answer right?

We probed 21 uniform depth percentiles (0%, 5%, 10%, ..., 100%) across all valid cells under a pre-registered **2-consecutive confirmation rule** ($\text{AUROC} \geq 0.65$ across two consecutive 5% taps). Activations were strictly tapped at the last prompt token position before generation.

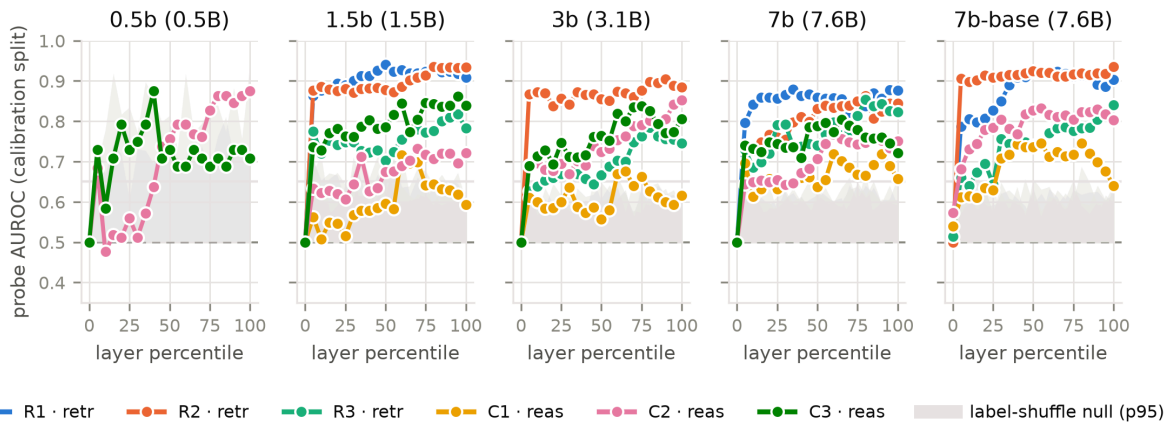


Figure 4: Continuous AUROC Emergence Across 21 Layer Taps. Retrieval tiers (cool colors) rise above the 0.65 threshold immediately at 5% depth. Reasoning tiers (warm colors) stay at chance for early layers and require deep computation before linear separability is achieved.

The Complete 24-Cell Distribution: Mean vs. Median

Evaluating the distribution of layer onsets across all 24 validated cells reveals an architectural dichotomy:

- **Factual Retrieval (\$N = 11\$):** [5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 15, 15]%
→ **Mean = 6.82%, Median = 5.0%.** Nine out of eleven cells onset at Layer 1 or 2!
- **Multi-Step Reasoning (\$N = 13\$):** [5, 5, 5, 5, 5, 15, 30, 40, 40, 45, 50, 60, 60]%
→ **Mean = 28.08%, Median = 30.0%.**
- **The Pre-Registered Test on Means:** Our primary pre-registered metric was the difference in mean onset: $\Delta_{\text{mean}} = +21.26\%$ [95% bootstrap CI: +9.72%, +33.05%]. A 20,000-draw non-parametric permutation test confirms statistical significance at **\$p = 0.0039\$**.
- **The Median Gap:** Complementarily, the median gap is $\Delta_{\text{median}} = +25.0\%$ (30.0% vs. 5.0%), demonstrating that the finding is robust to the Math floor effect.

Why Factual Memory is Instant, While Reasoning Takes Depth

The physical intuition here is beautiful:

- **Retrieval is an associative hash table lookup:** When you ask "*What is the capital of France?*", the query tokens pass through the initial token embeddings and first MLP block. The entity association is retrieved from weight storage almost instantaneously. The residual stream at Layer 1 already possesses a linearly separable representation of whether this fact was successfully retrieved.
- **Reasoning is iterative composition:** In GSM8K and GSM-Hard, solving the problem requires parsing variables, planning arithmetic operations, and routing intermediate results across attention heads. You cannot "look up" the solution to a novel word problem in Layer 1. In Qwen2.5-1.5B and 3B, internal correctness representations do not stabilize until **60% layer depth** (Layer 17/28 and Layer 22/36).

The MATH Floor Nuance: Why did 5 reasoning cells also onset at 5%? The data shows three competition MATH cells (1.5b__C3 , 3b__C3 , 7b-instruct__C3) and two algebraic reasoning cells (3b__C2 and 7b-base__C2 , one instruct and one base). On Olympiad math, accuracy was so dismal (<15%) that early probes keyed onto surface length and formatting tokens. In genuine multi-step arithmetic where models achieve meaningful accuracy (GSM8K and GSM-Hard), the reasoning ramp is unmistakable: [30, 40, 40, 45, 50, 60, 60]%. Even excluding GSM8K entirely, the gap remains **+12.63% [95% CI: +1.11%, +25.56%]**.

4. The "Swagger" Penalty: When Stated Confidence is Counter-Informative

What happens if you combine all three signals into a joint statistical model? We fitted a Bayesian multi-level logistic regression predicting correctness across 1,166 items in 12 multi-signal cells (with item random intercepts; note that individual per-cell fits degenerate due to collinearity, so this represents a pooled conditional association):

$$\text{logit}(P(\text{correct})) = \beta_0 + 7.16 \cdot \text{Behavioral} + 2.97 \cdot \text{Internal} - 3.26 \cdot \text{Verbal}$$

Notice the sign of the verbal coefficient: $\beta_{\text{Verbal}} = -3.26$ (SD = 0.25, $z = -13.0$).

This is the **Dunning-Kruger effect** formalized in generalized linear model coordinates. Once you hold behavioral consistency (sampling consensus) and internal representations constant, an LLM that still insists on stating high verbal confidence is *more likely to be wrong*. High stated confidence without underlying sample consensus is the mathematical signature of hallucination with swagger.

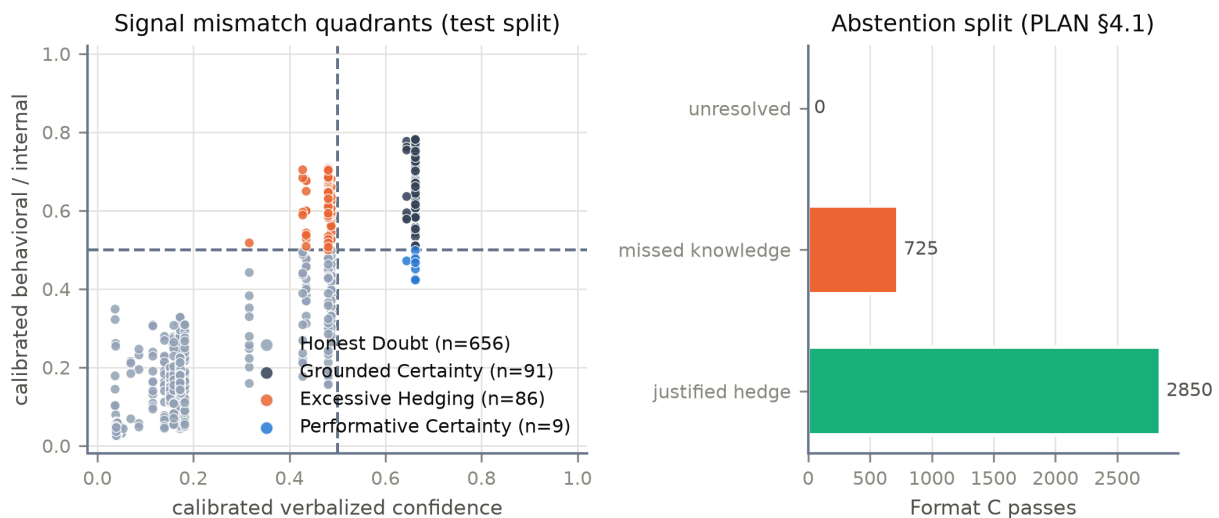


Figure 2: Epistemic Alignment Quadrants. Plotting calibrated verbal confidence against behavioral/internal consistency. The 10:1 ratio between Excessive Hedging and Performative Certainty was largely an artifact of the hardcoded 0.5 threshold. The continuous difference $\Delta = \text{Verbal} - \text{Behavioral}$ is symmetric around zero (-0.006).

5. The Confessions of a Pre-Registered Audit

The most important part of this research is what **didn't** work.

In contemporary AI research, there is immense pressure to smooth over rough edges, run fifty exploratory tests, and only report the ones that reached $p < 0.05$. We took the opposite approach: we pre-committed our hypotheses and audit gates in `PLAN.md` before final execution.

Here is the honest accounting of what the pre-registered audit caught:

1. **H0 (Verbal Format Invariance) was FALSIFIED:** We tested whether verbal confidence survives prompt formatting changes (Format A: 0–100%, Format B: qualitative verbal bins, Format C: betting stakes). The correlations collapsed to ρ in $[-0.07, +0.25]$ against a pre-registered bar of 0.60 . Verbal confidence is heavily prompt-dependent.
2. **H1 (Pre-Registered ECE Hierarchy) was FALSIFIED:** We predicted verbal confidence would have worse ECE than internal and behavioral signals. It didn't: pooled ECE was indistinguishable ($\Delta = 0.007$, CI overlapping zero). Only the post-hoc Murphy decomposition revealed the true resolution dominance.
3. **The Negative Control (p_0) Fired and Dropped a Cell:** We enforced a label-shuffled negative control ($p_0 \approx 0.50 \pm 0.10$). In `7b-base__c3`, the negative control scored $p_0 = 0.634$, signaling that the probe was fitting to token position artifacts. The pipeline automatically caught this and dropped the cell.
4. **Interaction Term Instability Disclosed:** In our baseline coarse run, the interaction term $\log(\text{params}) \times \text{Internal}$ was $+0.71$. When our fine 5% sweep rescued two cells at 60% depth and reduced imputation, that interaction term collapsed to $+0.03$. Rather than hiding this, we publish both model fits side-by-side in the paper.

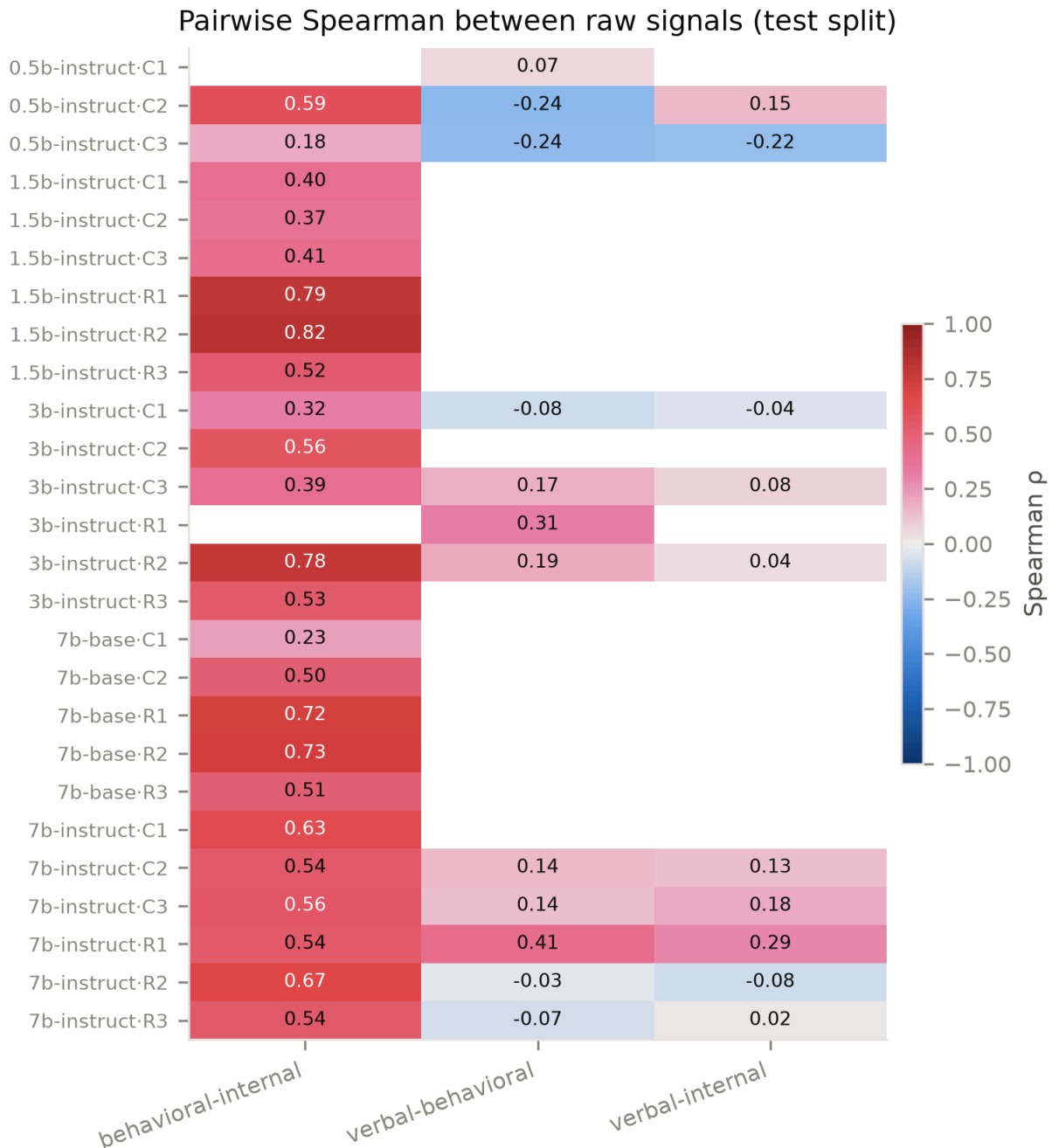


Figure 6: Cross-Signal Rank Correlation Matrix. Rank correlations between verbal, behavioral, and internal confidence across all 30 cells. Verbal confidence correlates weakly with internal hidden states ($\rho \approx 0.086$), proving persistent decoupling.

6. Practical Rules of Thumb for AI Safety Engineers

If you are designing automated routing, selective classification, or monitoring systems for LLMs in production, our results suggest three concrete design principles:

1. **Never Route on Verbalized Self-Confidence:** Asking an LLM "How sure are you?" is brittle, easily manipulated by prompt phrasing, and carries zero or negative partial correlation with correctness once consistency is controlled.
2. **Triage Factual Knowledge at Early Hidden Layers:** For knowledge-intensive QA, factual retrieval linearizes by Layer 1 or 2 (5% depth). You can train lightweight linear probes on early residual streams to reject unsupported queries before generating full output tokens.
3. **Pay the Compute Cost for Semantic Entropy on Complex Reasoning:** If the application involves multi-step logic or mathematics, internal probes at early layers are blind. Generating 5–10 rollouts and measuring semantic clustering provides over **2× the discriminative sorting resolution** of any single-pass signal.

Certainty in a neural network is not an afterthought written in words—it is a physical computation that takes time and depth to unfold.

The complete dataset, 21-tap activation tensors, and reproducible pipeline notebooks are open-sourced under the Confidence-AI benchmark repository bound to git commit SHAs `7570841` (baseline) and `a4608a8` (fine sweep).